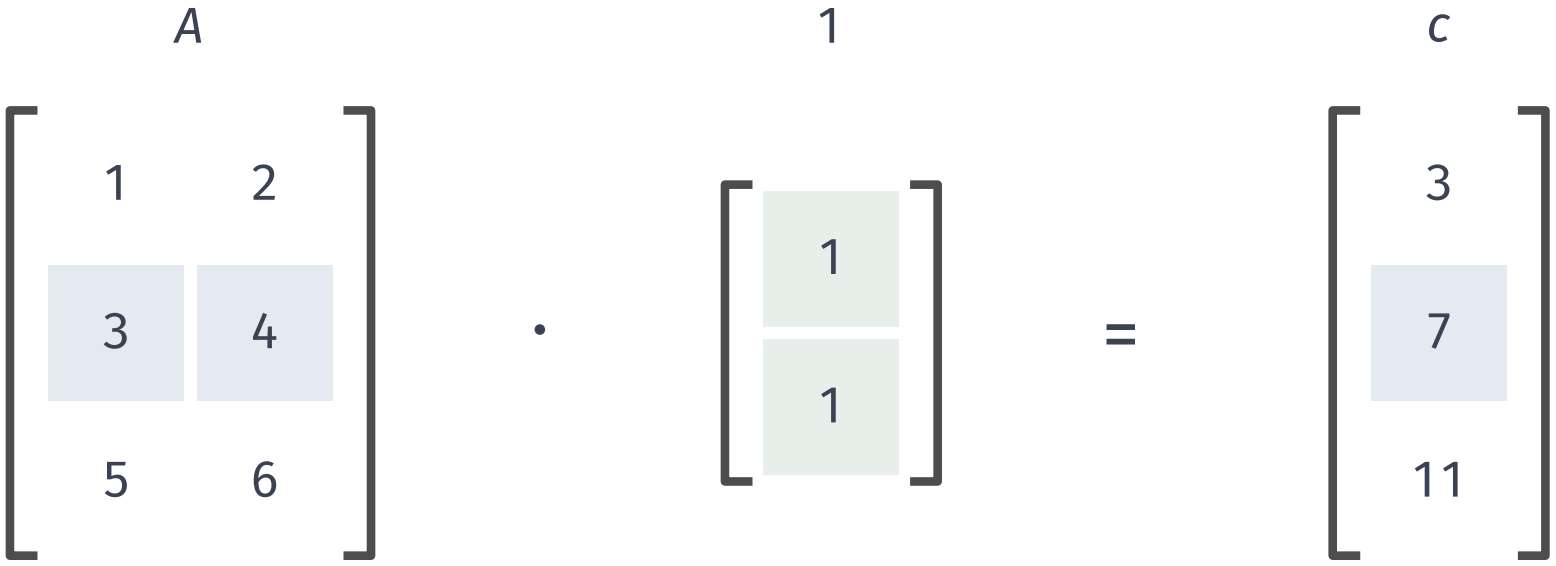

$$c_i = \sum_{j=1}^{2} A_{ij}\, 1_j$$